Chapter

Good Data Beats More Data: Building a Thermal Air-Image Dataset for Ground-to-Air Surveillance

Tijeni Delleji, Feten Slimeni and Ahmed Siala

Abstract

The efficiency of transfer learning for detecting drones using thermal imagery is highly dependent on the quality and diversity of the underlying training datasets. This chapter presents a comprehensive and reproducible pipeline for constructing high-quality aerial thermal datasets optimized for transfer learning applications. We detail the end-to-end process, encompassing thermal image acquisition, precise annotation, domain-specific augmentation, and dataset refinement through advanced cleaning techniques. To ensure robust model transferability across real-world scenarios, we emphasize the inclusion of environmental diversity—spanning diurnal cycles, weather variability, and heterogeneous landscapes such as urban and rural areas. We introduce novel thermal-specific augmentation strategies, including thermal blur, noise injection, and dynamic brightness modulation, to bridge the domain gap between pretraining and target deployment. Additionally, cleaning procedures using perceptual hashing and statistical outlier detection are employed to maintain dataset integrity and eliminate redundancy. Experimental evaluations confirm that datasets processed with our methodology yield superior performance in drone detection tasks, significantly enhancing accuracy, robustness, and generalization across varying conditions. This chapter thus provides practical insights and guidelines for researchers and practitioners aiming to leverage transfer learning in thermal imaging contexts, contributing to more reliable and scalable drone detection systems for ground-to-air surveillance.

Keywords: thermal imaging, ground-to-air surveillance, drone detection, dataset construction, data annotation, data cleaning, deep learning, infrared object detection

1. Introduction

The rapid proliferation of mini-drones has introduced a host of new security challenges. These compact and highly maneuverable aerial devices are increasingly being exploited for unauthorized surveillance, smuggling, and potential weaponization. As a result, their detection has become a critical concern for military, governmental, and civilian infrastructures alike [1, 2].

Real-time drone detection is especially vital in sensitive environments such as military bases, no-fly zones, and critical infrastructure, where any breach can have severe consequences [3–5]. Developing robust detection systems that can accurately identify and track mini-drones in diverse operational conditions has, therefore, become a top priority.

In this context, thermal imaging has emerged as a powerful modality for aerial surveillance. Unlike conventional optical cameras that rely on visible light, thermal imaging systems capture infrared radiation emitted by objects, allowing drones to be identified by their heat signatures. This capability is particularly advantageous in low-light or visually obstructed environments—such as during night operations or under conditions involving smoke, fog, or camouflage [6–10].

Thermal air imagery—capturing scenes from a ground-to-air perspective—has thus become a valuable resource for drone detection systems. However, this modality also introduces new challenges. Mini-drones often share thermal characteristics with other heat-emitting objects like birds, debris, or background noise, making accurate identification a non-trivial task, particularly for fast-moving and small targets.

To address these challenges, recent advances in deep learning—particularly object detection frameworks such as the YOLO (You Only Look Once) series—have shown promise in real-time thermal detection [11–13]. Modern YOLO architectures, including YOLOv8 through YOLOv11, have significantly improved in detection speed and precision [14–17]. Nevertheless, the performance of these models is highly influenced by the quality of the training dataset.

In this context, transfer learning plays a pivotal role, allowing pre-trained models to be adapted to targeted datasets (e.g., thermal drone imagery), significantly reducing training time and data requirements while enhancing accuracy. Yet, the success of transfer learning is contingent on the availability of clean, diverse, and well-annotated datasets that accurately reflect the target deployment environment.

In this chapter, we introduce a practical and systematic pipeline for constructing optimized thermal aerial datasets, purpose-built for transfer learning applications in drone detection. We cover key stages including thermal image acquisition across diverse conditions, precise data annotation, domain-specific augmentation (e.g., thermal blur, noise injection), and rigorous data cleaning using perceptual hashing and outlier detection.

By following this pipeline, researchers and practitioners can significantly enhance the generalization and robustness of drone detection models fine-tuned on thermal imagery. The chapter concludes with experimental insights, demonstrating how improved dataset construction strategies lead to measurable gains in detection accuracy and cross-condition adaptability.

The following sections of this chapter are organized as follows: Section 2 reviews relevant literature and recent advancements in thermal drone detection. Section 3 describes the dataset creation process, covering data collection, annotation, augmentation, and preprocessing. Section 4 presents our experimental framework, analyzing dataset characteristics and evaluating model performance. Finally, Section 5 summarizes key findings and suggests future research directions for transfer learning in thermal drone detection.

2. Related work

The effectiveness of drone detection systems using thermal imaging depends on two critical factors: robust algorithmic frameworks—whether developed from scratch, through fine-tuning, or via transfer learning—and high-quality training datasets. While prior research has extensively explored algorithmic advancements, the construction of optimized thermal datasets remains underexplored, despite being a key determinant of model performance.

2.1 Thermal imaging in drone detection

Thermal imaging has become a fundamental tool for drone detection due to its proven resilience under low-visibility conditions. Unlike optical sensors, thermal cameras capture infrared radiation, allowing for the reliable identification of drone heat signatures even in challenging scenarios such as darkness, fog, or smoke [8]. Early studies, such as Quan et al. [18], demonstrated the feasibility of using convolutional neural networks (CNNs) to distinguish drones from birds in thermal images. More recently, Soeleman et al. [10] integrated deep learning models with thermal data, achieving improved accuracy in cluttered environments. However, these studies have primarily focused on algorithmic enhancements, with limited emphasis on dataset quality, environmental diversity, and standardization—factors that are equally critical for achieving robust and generalizable performance.

2.2 Challenges in thermal dataset construction

Most existing drone detection datasets are based on RGB imagery [6], making them ill-suited for thermal-based applications due to sensor noise, low contrast, and thermal crossover effects. Publicly available thermal datasets, such as FLIR ADAS [19] and KAIST Multispectral [20], provide thermal-RGB paired data but lack dedicated aerial drone annotations. Additionally, custom datasets like [3] suffer from a limited scale (<5000 images) and fail to incorporate diverse environmental conditions, reducing their applicability to real-world surveillance.

Key limitations observed in prior thermal drone detection datasets:

- *Sensitivity to diurnal and seasonal variations*: Thermal signatures can fluctuate significantly depending on the time of day and season, impacting detection consistency.
- *Lack of multi-altitude perspectives*: Most datasets fail to capture data from varying flight altitudes, which is crucial for robust drone detection across different operational scenarios.
- *Presence of adversarial environmental conditions*: Challenging factors such as *solar glare*, *precipitation*, and *complex thermal backgrounds* are often underrepresented, limiting real-world applicability.

2.3 Algorithm: Dataset synergy for drone detection

The evolution of *YOLO-based architectures* (YOLOv8–YOLOv11) has significantly enhanced *small-object detection capabilities* [11, 15, 16]. However, these advancements often rely on access to *high-quality, well-curated datasets*. Delleji et al. [7] demonstrated that the *precision of YOLO models dropped significantly* when trained on *non-augmented imaging datasets*, emphasizing the strong *interdependence between dataset quality and model performance*. This underlines the *necessity of a holistic approach* that integrates *dataset curation, data cleaning, augmentation strategies, and real-world variability*—an aspect largely overlooked in prior literature.

Our work *addresses these gaps* by proposing a *standardized pipeline for thermal dataset creation*, incorporating:

- *Diverse environmental conditions* to enhance generalization across scenarios.
- *Data cleaning procedures* to eliminate mislabeled, corrupted, or low-quality samples that can hinder model performance.
- *Physics-based augmentations* to simulate real-world variability in thermal data.
- *Robust annotation validation protocols* to improve annotation quality and consistency.

2.4 Public thermal imaging datasets

Existing public thermal datasets primarily focus on *automotive, pedestrian, or general object detection*, with *limited emphasis on aerial drone detection*. Below is an overview of some of the most widely recognized thermal imaging datasets:

- *FLIR ADAS* [19]: A large-scale thermal dataset designed for *automotive applications*, featuring annotated thermal images of *vehicles, pedestrians, and cyclists*. Despite its size, it lacks *dedicated aerial drone annotations* and does not address challenges such as *thermal crossover at varying altitudes*.
- *KAIST multispectral* [20]: A benchmark dataset providing *aligned RGB-thermal image pairs*, primarily for *pedestrian detection*. However, its *annotations and scenarios are irrelevant* to drone detection and do not account for *thermal variations across different weather conditions or altitudes*.
- *LITIV dataset* [21]: Focused on *low-light and thermal object tracking*, this dataset includes *dynamic outdoor scenes* but lacks *systematic drone annotations* or consideration of *adversarial conditions* such as *solar glare or precipitation*.
- *VAP thermal aerial dataset* [22]: A small dataset containing *thermal aerial imagery of birds and UAVs*. However, it suffers from a *limited scale* (fewer than 2000 frames) and minimal *environmental diversity*, restricting its *usefulness for training robust drone detection models*.
- *ARL thermal dataset* [23]: Developed for *military applications*, this dataset includes thermal sequences of *vehicles and humans* in cluttered environments. While it addresses some *thermal crossover challenges*, it lacks *multi-altitude drone perspectives* and does not capture *seasonal variations*.

While these datasets provide foundational support for thermal imaging research, they fall short in systematically addressing the key requirements for aerial drone detection. This highlights the urgent need for a dedicated thermal drone detection benchmark that accounts for diverse environmental conditions, varying altitudes, and challenging real-world scenarios. To bridge this gap, our study introduces a comprehensive, well-annotated, and publicly available thermal aerial imagery dataset, specifically tailored for ground-to-air drone detection.

3. Construction pipeline of the thermal dataset

The development of our dataset, *Thermal Air-Image Dataset for Ground-to-Air Surveillance*, follows a well-defined and systematic pipeline comprising the following stages:

1. *Raw data collection*: Thermal imagery is captured using an EO/IR imaging system [24], ensuring high-quality input data suitable for surveillance applications.

2. *Data augmentation*: To improve diversity and robustness, the collected images undergo augmentation techniques such as the addition of blur and noise, simulating varied environmental and sensor conditions.

3. *Annotation:* Objects of interest within the thermal images are meticulously labeled to provide accurate and reliable ground-truth data for training and evaluation.

4. *Data cleaning:* The dataset is refined by removing duplicate entries and correcting inconsistencies, thereby preserving data integrity and consistency.

5. *Dataset partitioning*: The final dataset is systematically divided into training, validation, and test subsets, ensuring balanced distribution and effective model evaluation.

A more detailed description of each step is provided below.

3.1 Data collection

A crucial element for any deep learning-based system is acquiring high-quality training and validation data. Our data collection methodology employed a sophisticated Electro-Optical and Infrared (EO/IR) imaging system for thermal data acquisition. The system's central component, illustrated in **Figure 1**, consists of the multisensor PTZ camera (model DH-TPC-PT8621C, TPC 2024). This advanced surveillance equipment incorporates a dual-lens architecture that enables synchronous capture of both visible spectrum and thermal infrared video streams. This all-in-one solution, equipped with sophisticated sensors, is particularly effective for long-distance outdoor surveillance, ensuring comprehensive data collection under varying conditions and significantly improving the detection accuracy and operational reliability.

The system integrates an RGB camera equipped with starlight technology and a motorized zoom lens, enabling high-resolution imaging and detailed close-up verification. For low-visibility conditions (day or night), a dedicated thermal imaging camera detects emitted infrared radiation, ensuring reliable surveillance regardless of lighting. An integrated laser illuminator paired with the RGB camera enhances nighttime imaging, simulating reflected infrared capabilities.

Figure 1.
Multisensor network PTZ camera.

Table 1 presents the technical specifications of the thermal camera, including its VOX uncooled detector, 6.2° × 5.0° field of view, ≤ 40 mK sensitivity, 100 mm focal length, 8 μm to 14 μm spectral range, 640 × 512 resolution, 17 μm pixel pitch, and 25 fps frame rate.

This study utilized the imaging system to capture thermal videos of the three commercial DJI mini-UAVs listed in **Table 2**: the Phantom 4 Pro+ V2.0, Matrice 600, and Mavic 3. Our data collection strategy involved recording video sequences during diverse daylight conditions, varying illumination levels, and multiple challenging settings to accurately replicate operational environments. The thermal recordings, taken at 25 frames per second (FPS), were saved in MP4 format. From these recordings, a selection of key images was extracted to form the dataset's foundation, ensuring exposure to diverse thermal signatures and environmental conditions during model training.

3.2 Enhancing dataset robustness *via* thermal data augmentation

Thermal data augmentation is a critical technique used to enhance the performance of neural network models in thermal imaging by artificially increasing the

Parameter	Specification
Detector type	VOX uncooled focal plane detector
Field of view	6.2° × 5.0°
Thermal sensitivity (NETD)	≤ 40 mK
Focal length	100 mm
Spectral range	8 – 14 μm (LWIR)
IR resolution	640 × 512
Pixel pitch	17 μm
Video frame rate	25fps

Table 1.
Technical characterization of the Thermal Camera.

Parameter	DJI Mavic 3	DJI Phantom 4 Pro + V2.0	DJI Matrice 600
Size	8.3 × 8.3 × 3.3 cm	19.6 × 29 × 29 cm	166.8 × 151.8 × 75.9 cm
Weight	730 g	1375 g	9100 g
Diagonal Length	38.1 cm (15 in)	35.6 cm (14 in)	113.3 cm (44.6 in)
Payload	Camera (up to 1 kg)	Camera (up to 1.5 kg)	6 kg (cameras, LiDAR, etc.)
Max flight time	46 minutes	30 minutes	35 minutes
Max speed	75 kph (47 mph)	72 kph (45 mph)	65 kph (40 mph)
Max range	15 km (9.3 mi)	7 km (4.3 mi)	5 km (3.1 mi)
Application	Photography, cinematography	Photography, mapping	Industrial, surveying, cinematography

Table 2.
Technical characterization of the considered min-UAVs.

diversity of the training data without the need to collect new samples. Given the unique characteristics of thermal images, data augmentation is vital in ensuring that the model is robust against various environmental and operational variations. By expanding the training set with new, plausible examples, the detection model becomes better equipped to handle the complexities of real-world scenarios when deployed.

In this study, we applied several thermal-specific data augmentation techniques using the online computer vision development framework, Roboflow [25]. These techniques include:

1. *Thermal BLUR*: Introducing a blur effect to thermal images helps the model develop resistance to variations in camera focus. This can be mathematically represented as the convolution of the thermal image $I(x,y)$ with a Gaussian kernel $G(x,,y,,\sigma)$:

$$I_{\text{blur}}(x,y) = I(x,y) * G(x,y,\sigma) \tag{1}$$

Which is defined as:

$$G(x,y,\sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{2}$$

In this equation, σ controls the standard deviation, influencing the degree of blurriness introduced to the image.

2. *Temperature brightness variability*: Adjusting the brightness in thermal images simulates different temperature conditions, enabling the model to be more robust to changes in environmental temperature and camera settings. This can be modeled as:

$$I_{\text{brightness}}(x,y) = I(x,y) + \Delta T \cdot k \tag{3}$$

where ΔT is a constant representing the temperature change, and k is a scaling factor that adjusts the intensity of the brightness modification based on the pixel's temperature.

3. *Thermal noise*: Adding noise to thermal images mimics the artifacts and distortions often encountered in thermal imaging due to sensor imperfections or environmental factors. The noise can be represented by:

$$I_{\text{noise}}(x,y) = I(x,y) + N(x,y) \tag{4}$$

where $N(x,y)$ is a noise function drawn from a Gaussian distribution $N(0,\sigma_n^2)$ defined as:

$$N(x,y) \sim \mathcal{N}(0,\sigma_n^2) \quad \text{for}(x,y) \in \Omega \tag{5}$$

Here, σ_n stands for the standard deviation of the noise, and Ω for the domain of the image.

4. *Thermal CutMix*: Similar to the traditional CutMix, this technique involves randomly cutting out a portion of the thermal image and replacing it with a segment from another image. The CutMix operation can be mathematically expressed as follows:

$$I_{mix}(x,y) = \begin{cases} \lambda I_1(x,y) + (1-\lambda) I_2(x',y'), \\ \quad \text{if}(x,y) \in \text{CutOutRegion}, \\ I_1(x,y), & \text{otherwise.} \end{cases} \tag{6}$$

Where (x',y') represent the coordinates of the cutout region in I_2, λ is a mixing parameter that controls the contribution of each image to the final output, and *CutOutRegion* refers to the portion of the original image that is replaced.

By applying these targeted augmentation techniques, we significantly improved the model's robustness and accuracy in detecting objects within thermal datasets. The enhanced diversity and variability in the training data prepare the model for deployment in diverse and challenging thermal imaging environments, allowing it to adapt effectively to real-world scenarios.

3.3 Thermal data annotation

Data annotation is a crucial process in deep learning (DL) applications, particularly for tasks involving identifying and categorizing objects based on their heat signatures in thermal images. This specialized form of annotation involves drawing bounding boxes, polygons, or other shapes around objects within thermal photos and assigning them specific labels, such as "drone," "bird," or "plane." Unlike traditional image annotation, which relies on visible light, thermal image annotation focuses on the infrared spectrum, where objects are detected based on the heat they emit. The accuracy and consistency of this annotation process are paramount, as the performance of DL models is highly dependent on the quality of the annotated data.

Reliable annotation requires a deep understanding of the unique thermal patterns associated with different objects and the ability to discern subtle temperature variations. Additionally, it is essential to ensure that only relevant features are annotated, excluding any thermal artifacts that do not belong to the target object classes. By meticulously labeling thermal data, annotators contribute to creating robust datasets critical for training DL models capable of accurately detecting and identifying objects in complex and challenging environments.

3.4 Thermal data cleaning

In optimizing our object detection model, we embraced the principle that "Good data beats more data," as emphasized by Dedrone [26], emphasizing the quality and diversity of images in our dataset. This approach was especially crucial in thermal data, where image quality can significantly impact model performance. After meticulous data annotation, we undertook a thorough refinement process to ensure the integrity of our thermal dataset, which involved identifying and resolving issues such as conflicting labels, data duplicates, label correlation, and high filter correlation. A significant challenge we encountered was the presence of near-duplicates—visually similar images but not identical—which can lead to redundancy and biased results in machine learning datasets.

We employed perceptual hashing techniques specifically tailored for thermal images to tackle this. Perceptual hashing generates a hash based on the visual content of images, allowing for the detection of near-duplicates that exhibit subtle variations, such as slight changes in temperature gradients. We utilized the phash algorithm with a hash size of 8, which produces a compact representation of each image's visual content. The process begins by applying the Discrete Cosine Transform (DCT) to convert the image into frequency components, enabling us to capture the essential features of the image while disregarding less relevant details. The DCT of an image I can be expressed mathematically as:

$$DCT(u,v) = \frac{1}{4}C(u)C(v)\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} I(x,y)\cos\left(\frac{(2x+1)u\pi}{2N}\right) \times \cos\left(\frac{(2y+1)v\pi}{2N}\right) \tag{7}$$

where $C(u)$ and $C(v)$ are normalization factors defined as:

$$C(u) = \begin{cases} \frac{1}{\sqrt{2}}, & \text{if } u = 0, \\ 1, & \text{otherwise.} \end{cases} \tag{8}$$

After computing the DCT coefficients, we derive a binary hash for each image by comparing these coefficients to the median value of the DCT array. The binary hash b is constructed as follows:

$$b_{i,j} = \begin{cases} 1, & \text{if } DCT(i,j) \geq \text{median}(DCT), \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

This results in a binary representation that succinctly captures the visual features of the image. To identify near-duplicates, we compare the binary hashes of different images. The Hamming distance d between two binary hashes b_1 and b_2 is calculated as:

$$d(b_1, b_2) = \sum_{i=1}^{n} \left| b_1[i] - b_2[i] \right| \tag{10}$$

where n is the length of the hash. If the Hamming distance d between two hashes falls below a predefined threshold T, we classify those images as near-duplicates:

$$\text{if } d(b_1, b_2) < T, \text{ then images are near-duplicates} \tag{11}$$

By efficiently identifying and removing these near-duplicates, we streamline our dataset, reducing redundancy and ensuring that each image contributes meaningfully to the training of our model. Overall, this phase-based detection of near-duplicates has proven to be a valuable tool for cleaning and organizing high-quality datasets, enabling us to enhance the performance of our thermal image object detection model.

3.5 Dataset splitting

Strategic dataset partitioning is crucial for developing models capable of generalizing to unseen data, with careful separation between training, validation, and test subsets. We curated 6000 sample images from the collected thermal videos, called "Seen data." To create a robust training pipeline, these images were randomly divided into training and validation sets, with 4800 images (80%) allocated for training and 1200 images (20%) reserved for validation. This split was carefully chosen to maintain a balanced distribution, ensuring that the model receives a broad exposure during training while allowing for meaningful performance evaluation during validation.

Additionally, 600 images were extracted from videos that were not part of the training and validation sets to assess the model's performance on entirely new scenarios. Following standard evaluation protocols, these images were reserved as an independent test set, explicitly labeled "unseen data" to simulate real-world deployment conditions. The splitting strategy is visually summarized in **Figure 2**, which illustrates the proportion of data allocated to training, validation, and testing. The "Seen data" (training and validation) is depicted with an 80% training set and a 20% validation set. In contrast, the "Unseen data" forms the testing set, ensuring a clear distinction between the datasets.

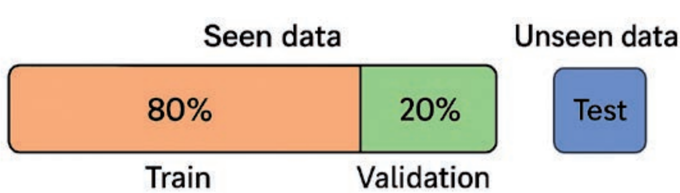

Figure 2.
Cross-thermal dataset: training, validation (seen data), and test (unseen data) sets.

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The entire dataset, encompassing training, validation, and testing sets, was combined into a comprehensive "drone thermal dataset," which will be publicly available for future research endeavors.

4. Experiments

4.1 Dataset exploration

Figure 3 displays a pair plot, illustrating the relationships between four variables commonly used in object detection tasks: x, y, width, and height of bounding boxes. The diagonal elements show histograms for each variable, revealing their distributions, with x and y appearing uniformly distributed, while width and height show skewed distributions toward smaller values. The off-diagonal scatter plots represent the pairwise relationships between these variables, with notable patterns such as a positive correlation between width and height. In contrast, other variable pairs exhibit more scattered distributions without clear correlations. This visualization aids in understanding the distribution and interdependencies of bounding box parameters in a dataset.

Figure 4 provides a detailed visualization of the thermal dataset labels. The top-left plot shows a label count for the "Drone" class, indicating approximately 2500 instances, while the top-right visualizes the overlap and distribution of bounding boxes, showing

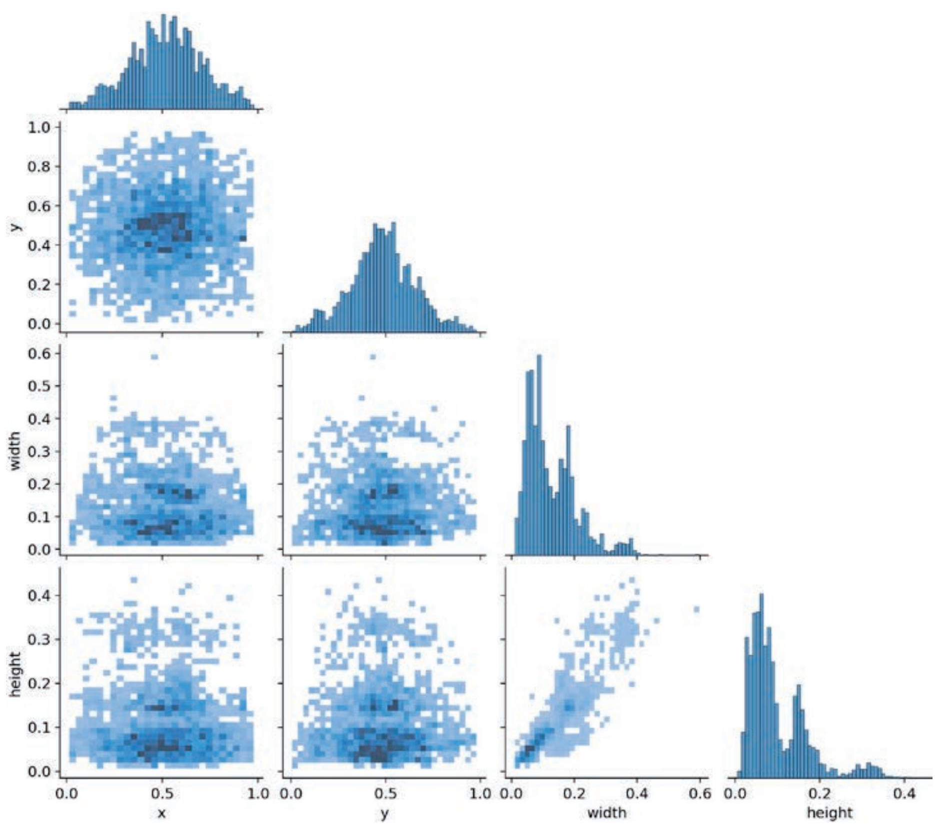

Figure 3.
correlogram of drone label dimensions: height vs. width correlation.

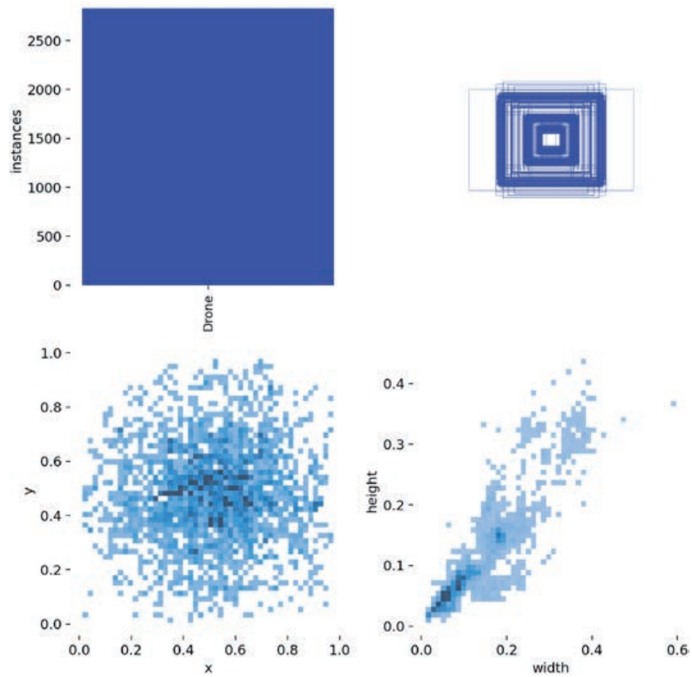

Figure 4.
Label volume and distribution for drones detection in our customized dataset.

how objects are positioned and sized within the images. The scatter plots below illustrate the relationships between bounding box attributes, such as x *vs* y coordinates, which are evenly spread, and width *vs* height, showing a positive correlation, meaning larger width tends to coincide with more considerable height. Overall, this figure offers insights into the spatial distribution and size variation of the bounding boxes within the dataset.

Figure 5 illustrates sample thermal air-images from the dataset, showcasing diversity across different lighting conditions, drone models, and environmental backgroundsthe diversity of thermal air-images within the customized dataset. Our dataset encompasses a wide range of ground-to-air scenarios, capturing various environmental conditions, altitudes, and object types. This diversity enhances its applicability for robust model training and real-world surveillance applications.

4.2 Performance analysis

This section evaluates the impact of data cleaning on model performance by analyzing dataset distribution and key evaluation metrics.

First, **Table 3** presents a comparison of the dataset size before and after the cleaning process. The number of images in both the training and validation sets decreases after cleaning, indicating the removal of potentially replicate or irrelevant data. This step ensures a more refined dataset, which can lead to improved generalization and model robustness.

Next, **Figure 6** illustrates the effect of data cleaning on precision across training steps using the YOLOv11 [17] model for object detection. The plotted curves for "Before Cleaning" and "After Cleaning" indicate similar precision, despite the dataset

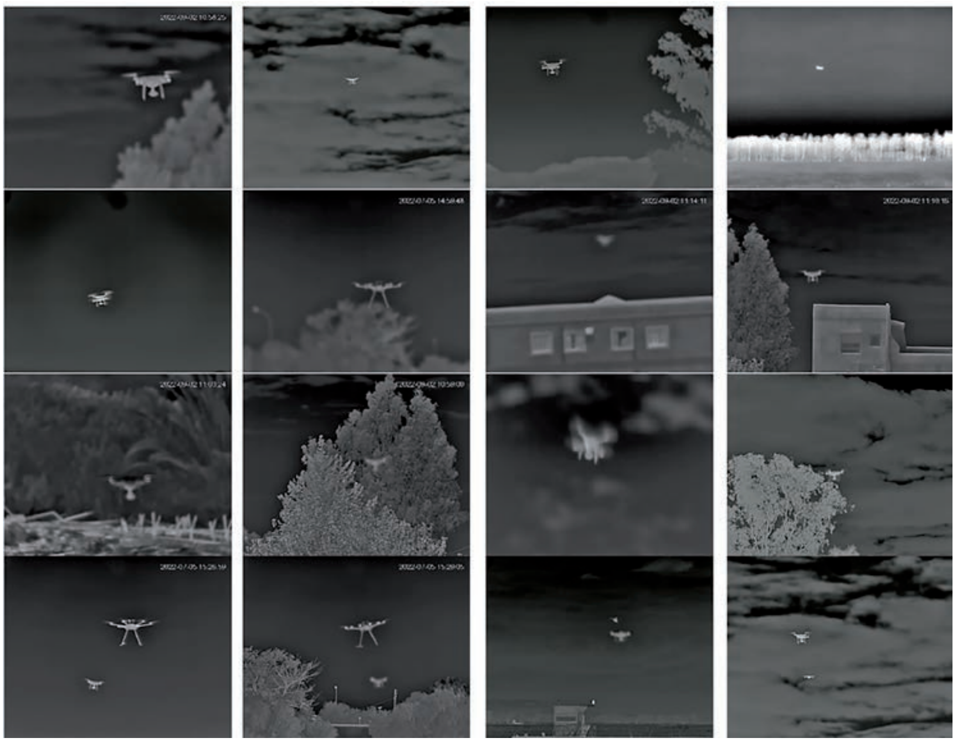

Figure 5.
Sample thermal air-images from the dataset, showcasing diversity across different lighting conditions, drone models, and environmental backgrounds.

	Images number	Training set	Validation set
Before Cleaning	12,600	9600	3000
After Cleaning	10,200	7600	2600

Table 3.
Dataset split before and after cleaning.

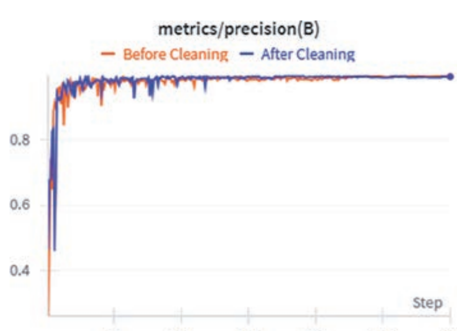

Figure 6.
Effect of data cleaning on precision across training steps in drone detection using YOLOv11.

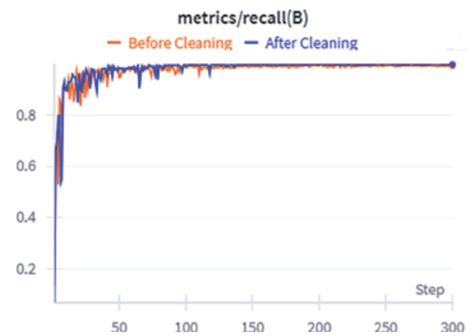

Figure 7.
Impact of data cleaning on recall performance in drone detection using YOLOv11.

before cleaning being larger (12,600 images) compared to after cleaning (10,200 images). This suggests that data cleaning effectively removes redundant or irrelevant samples without compromising the model's precision.

Similarly, **Figure 7** (recall analysis) compares the recall performance of the YOLOv11 model [17] before and after data cleaning. The recall metric is essential for determining how well the model captures true positive instances. Despite the cleaned dataset being smaller than the original, the recall remains similar, suggesting that the removed data had minimal impact on the model's ability to detect positive cases. This indicates that data cleaning effectively eliminates redundant or noisy samples without compromising recall performance.

Overall, this analysis provides insights into the effectiveness of dataset cleaning in refining the dataset by removing noisy or redundant samples while maintaining model precision and recall, thereby ensuring accuracy and consistency.

5. Conclusion

This study presents a systematic framework for constructing high-quality thermal air-image datasets to enhance drone detection in ground-to-air surveillance applications. By integrating diverse environmental conditions, rigorous quality control measures, and advanced augmentation techniques, the proposed dataset significantly improves detection accuracy, model robustness, and generalization. Key contributions include a comprehensive dataset pipeline covering data acquisition, annotation, filtering, and augmentation, along with a publicly available dataset designed to enhance real-world drone detection. Additionally, insights into optimizing thermal data for small-object detection are provided, ensuring its applicability in complex surveillance scenarios. Future research directions will involve: (1) dataset expansion incorporating diverse drone platforms, (2) performance evaluation under adversarial environmental conditions, and (3) integration of multi-sensor fusion approaches to enhance real-time detection robustness. This framework provides researchers with a comprehensive foundation for advancing thermal-based drone surveillance systems.

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Author details

Tijeni Delleji*†, Feten Slimeni† and Ahmed Siala†
Military Research Center, Aouina Military Base, Taieb Mhiri City, Tunis, Tunisia

*Address all correspondence to: tijeni_dalleji@yahoo.fr

† These authors contributed equally.

References

[1] Seidaliyeva U, Ilipbayeva L, Taissariyeva K, Smailov N, Matson ET. Advances and challenges in drone detection and classification techniques: A state-of-the-art review. Sensors. 2024;**24**(1):1-31. [Online]. Available from: https://www.mdpi.com/1424-8220/24/1/125

[2] Delleji T, Slimeni F, Lafi MA. C4 software for anti-drone system. Defence Science Journal. 2024;**74**(5):635-642

[3] Taha B, Shoufan A. Machine learning-based drone detection and classification: State-of-the-art in research. IEEE Access. 2019;**7**:138669-138682

[4] Ramadan R, Emara AH, Al-Sarem M, Elhamahmy M. Internet of drones intrusion detection using deep learning. Electronics. 2021;**10**:10

[5] Tu M, Le K, Phung MD. Object detection in thermal images using deep learning for unmanned aerial vehicles. Computer Vision and Pattern Recognition. 2024;**01**:687-692

[6] Delleji T, Fkih H, Kallel A, Chtourou Z. Visual tracking of mini-UAVs using modified YOLOv5 and improved DeepSORT algorithms. In: Proceedings of the 2022 6th International Conference on Advanced Technologies for Signal and Image Processing (ATSIP). Piscataway, NJ, USA: IEEE; 2022. pp. 1-6

[7] Delleji T, Chtourou Z. An improved YOLOv5 for real-time mini-UAV detection in no-fly zones. In: Proceedings of the 2nd International Conference on Image Processing and Vision Engineering (IMPROVE). Setúbal, Portugal: SciTePress; 2022. pp. 174-181

[8] Ding B, Zhang Y, Ma S. A lightweight real-time infrared object detection model based on YOLOv8 for unmanned aerial vehicles. Drones. 2024;**8**(9):1-22. [Online]. Available from: https://www.mdpi.com/2504-446X/8/9/479

[9] Quan CY, Edmond OLW, Srigrarom S. Identification of drone thermal signature by convolutional neural network. In: Proceedings of the 2021 International Conference on Unmanned Aircraft Systems (ICUAS). Vol. 2021. Piscataway, NJ, USA: IEEE; 2021. pp. 63-70

[10] Soeleman M, Supriyanto C, Purwanto. Deep learning model for unmanned aerial vehicle-based object detection on thermal images. Revue d'Intelligence Artificielle. 2023;**37**(6):1441-1447

[11] Delleji T, Slimeni F, Lafi M, Ayadi A, Chtourou Z. Deep sky monitoring system for mini-drone detection and tracking. In: 2023 IEEE International Conference on Advanced Systems and Emergent Technologies (IC_ASET). 2023. pp. 1-6

[12] Delleji T, Slimeni F, Lafi M, Chtourou Z. Video data analytics dashboard for anti-drone system. In: Proceedings of the 2023 International Conference on Cyberworlds (CW), Sousse, Tunisia. Piscataway, NJ, USA: IEEE; 2023. pp. 134-140

[13] Khan MA, Menouar H, Eldeeb A, Abu-Dayya A, Salim FD. On the detection of unauthorized drones—Techniques and future perspectives: A review. IEEE Sensors Journal. 2022;**22**(12):11439-11455

[14] Delleji T, Slimeni F. Rf-yolo: A modified yolo model for uav detection and classification using rf spectrogram images. Telecommunication Systems. 2025;**88**(1):33. DOI: 10.1007/s11235-025-01264-4

[15] Wang C-Y, Yeh I-H, Liao H. Yolov9: Learning what you want to learn using programmable gradient information. ArXiv: abs/2402.13616. 2024. [Online]. Available from: https://api.semanticscholar.org/CorpusID:267770251

[16] Wang A, Chen H, Liu L, Chen K, Lin Z, Han J, et al. Yolov10: Real-time end-to-end object detection. arXiv preprint arXiv:2405.14458. 2024

[17] Khanam R, Hussain M. Yolov11: An overview of the key architectural enhancements. arXiv preprint. October 2024

[18] Quan CY, Wong EO, Srigrarom S. Identification of drone thermal signature by convolutional neural network. In: Proceedings of the 2021 International Conference on Unmanned Aircraft Systems (ICUAS), Athens, Greece. Piscataway, NJ, USA: IEEE; 2021. pp. 63-70

[19] I. FLIR Systems. Flir thermal dataset for automotive applications. 2019. Available from: https://www.flir.com [Accessed: February 14, 2025]

[20] Hwang S, Park J, Kim N, Choi Y, Kweon IS. Multispectral pedestrian detection: Benchmark dataset and baseline. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Boston, MA, USA: IEEE; Jun 2015. pp. 1037-1045

[21] Gade R, Moeslund TB. Thermal cameras and applications: A survey. Machine Vision and Applications. Jan 2014;**25**(1):245-262. DOI: 10.1007/s00138_013_0570_5

[22] Berg A et al. Thermal aerial detection using deep learning. In: IEEE International Conference on Robotics and Automation (ICRA). 2018

[23] Jacobs N et al. The ARL multimodal dataset for thermal and visible object detection. In: Proceedings of SPIE Defense + Security. Vol. 9841. Bellingham, WA, USA: SPIE; 2016. DOI: 10.1117/12.2223089

[24] Dahua technology. 2024. Aailable from: https://www.dahuasecurity.com/products/All-Products/Thermal-Cameras/Wizmind-Series/TPC-8-Series/TPC-PT8621C/ [Accessed: September 04, 2024]

[25] Roboflow. Roboflow: Computer Vision Development Platform. 2023. Available from: https://roboflow.com

[26] Dedrone. How Dedrone Just Delivered Its Fastest and Most Accurate Airspace Security Solution Yet. 2024. Available from: https://www.dedrone.com/blog/how-dedrone-just-delivered-its-/fastest-and-most-accurate-/airspace-security-solution-yet [Accessed: August 21, 2024]